UNIVERSITY OF PRINCE EDWARD ISLAND

STAT 1210-2 — Introduction to Statistics for the Life and Social Sciences

Fall 2026

School of Mathematical and Computational Sciences

Instructor Information

Instructor: Ali Raisolsadat
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Office Hours: Cass Science Hall, Room 105 (SMCS Help Centre); Tuesdays and Thursdays, 2:15 PM–3:30 PM
Lectures: Tuesdays and Thursdays, 1:00 PM–2:15 PM
Classroom: Irving Chemistry Centre, Room 104
Course Website: `https://ali-raisolsadat.github.io/courses`

Moodle will be used for official announcements, communication, schedule updates and grades.

Course Description

STAT 1210 is an introductory statistics course for students in the life and social sciences. The course introduces descriptive statistics, probability, sampling, statistical inference, correlation and introductory regression. Emphasis will be placed on statistical reasoning, interpretation and the appropriate use of statistical methods in applications from areas such as biology, health sciences, psychology, environmental science and economics.

Prerequisite

Grade XII academic Mathematics. Previous experience with statistics or programming is not required.

Textbook

The Practice of Statistics in the Life Sciences, Brigitte Baldi and David S. Moore.

Course Materials

Lecture slides, course notes, assignments, practice material and other supporting resources will be made available through the course website and/or Moodle. Students are responsible for checking Moodle regularly for announcements and schedule changes.

Course Content

I. Exploring Data

- **Chapters 1–2:** Visualization and Numerical Summaries
- **Chapter 3:** Scatterplots and Correlation
- **Chapter 4:** Regression

II. Probability and Distributions

- **Chapters 9–10:** Probability
- **Chapter 11:** Normal Distributions
- **Chapter 12:** Binomial and Poisson Distributions

III. Foundations of Statistical Inference

- **Chapter 6:** Samples and Observational Studies
- **Chapter 13:** Sampling Distributions
- **Chapter 14:** Confidence Intervals and p-values
- **Chapter 15:** Inference in Practice

IV. Specific Inference Procedures

- **Chapter 17:** Inference about a Population Mean
- **Chapter 19:** Inference about a Population Proportion

The exact pacing and amount of material covered may be adjusted during the semester when necessary.

Statistical Software

R may occasionally be used during lectures for statistical calculations, visualization, and data analysis. No previous programming experience is required.

Assessment and Grading

Assessment	Weight
Three Assignments	20%
Midterms or Quizzes	40%
Final Examination	40%
Total	**100%**

Assignments

There will be three assignments. Each assignment will include 7-12 questions and one or two optional bonus questions. If a student earns a score greater than 100% on an assignment by completing the bonus questions, **one additional percentage point** will be added to the student's quiz or midterm grade. This is a one-percentage-point increase to the applicable quiz or midterm mark, not an additional 1% toward the final course grade.

Midterm or Quiz Option

The examination component will account for 40% of the final grade. One of the following formats will be used:

- **Option A:** Two 75-minute midterm tests worth 20% each.
- **Option B:** Four 40-minute quizzes worth 10% each.

The selected format and associated dates will be confirmed early in the semester.

Final Examination

The final examination will be worth 40% of the final grade. The date and time will be determined according to the University's final examination schedule.

Class Cancellation

If a class is cancelled because of weather or another unexpected circumstance, course material and assessment dates may be adjusted as necessary. A quiz or midterm scheduled for a cancelled class will normall move to the next class unless otherwise announced.

Calculators and Electronic Devices

A non-programmable, non-graphing calculator may be used during quizzes and midterm tests unless otherwise announced. Cell phones, smart watches, tablets, laptops, translators and other electronic devices may not be used during quizzes, tests or examinations unless specifically authorized.

Academic Integrity

Students are expected to follow the University of Prince Edward Island's Academic Integrity regulations. Academic dishonesty includes plagiarism, cheating, unauthorized collaboration, falsification of information, unauthorized use of technology, and other forms of misrepresentation of academic work. Suspected cases of academic dishonesty will be addressed according to University procedures.

The complete regulations are available at:

`https://calendar.upei.ca/current/chapter/undergraduate-and-professional-programs-academic-regulations/`

Use of ChatGPT and Other Generative AI Tools

ChatGPT, Claude, Gemini and other generative AI tools may be used as learning aids unless the instructions for a particular assessment state otherwise. Students must not submit AI-generated answers as their own work or use these tools during quizzes, tests, or examinations unless explicitly authorized.

Students are responsible for understanding and verifying all submitted work. When submitted work raises concerns regarding unauthorized assistance, a student may be asked to explain the calculations, reasoning, interpretations or conclusions contained in the submission. Academic integrity concerns will be handled according to University procedures.

Tentative Course Schedule

The following schedule is tentative and may be adjusted during the semester. Any changes will be announced through Moodle.

Week (Mon–Fri)	Tuesday	Thursday	Deliverables
Sept 7–11	Chapter 0; Chapter 1	Chapter 1	—
Sept 14–18	Chapter 2	Chapter 3	Introduction Task Memo Due Sept 15 Assignment 1 Published Sept 15
Sept 21–25	Chapter 3; Chapter 4	Chapter 4	—
Sept 28–Oct 2	Quiz 1; Chapter 9	Chapter 9	Assignment 1 Due Sept 29 Assignment 2 Published Sept 29
Oct 5–9	Chapter 10	Chapter 10; Chapter 11	—
Oct 12–16	Mid-semester break	Mid-semester break	—
Oct 19–23	Chapter 11	Chapter 11; Chapter 12	—
Oct 26–30	Chapter 12	Quiz 2; Chapter 12	—
Nov 2–6	Chapter 6	Chapter 13	Assignment 2 Due Nov 3 Assignment 3 Published Nov 3
Nov 9–13	Chapter 14	Chapter 14	Nov 11: Remembrance Day (No classes)
Nov 16–20	Chapter 15	Chapter 15	—
Nov 23–27	Quiz 3; Chapter 17	Chapter 17	Assignment 3 Due Nov 23
Nov 30–Dec 4	Chapter 19	Quiz 4; Final Exam Review	—
Dec 7–11	Final Exam Review	Final Exam Review	—